

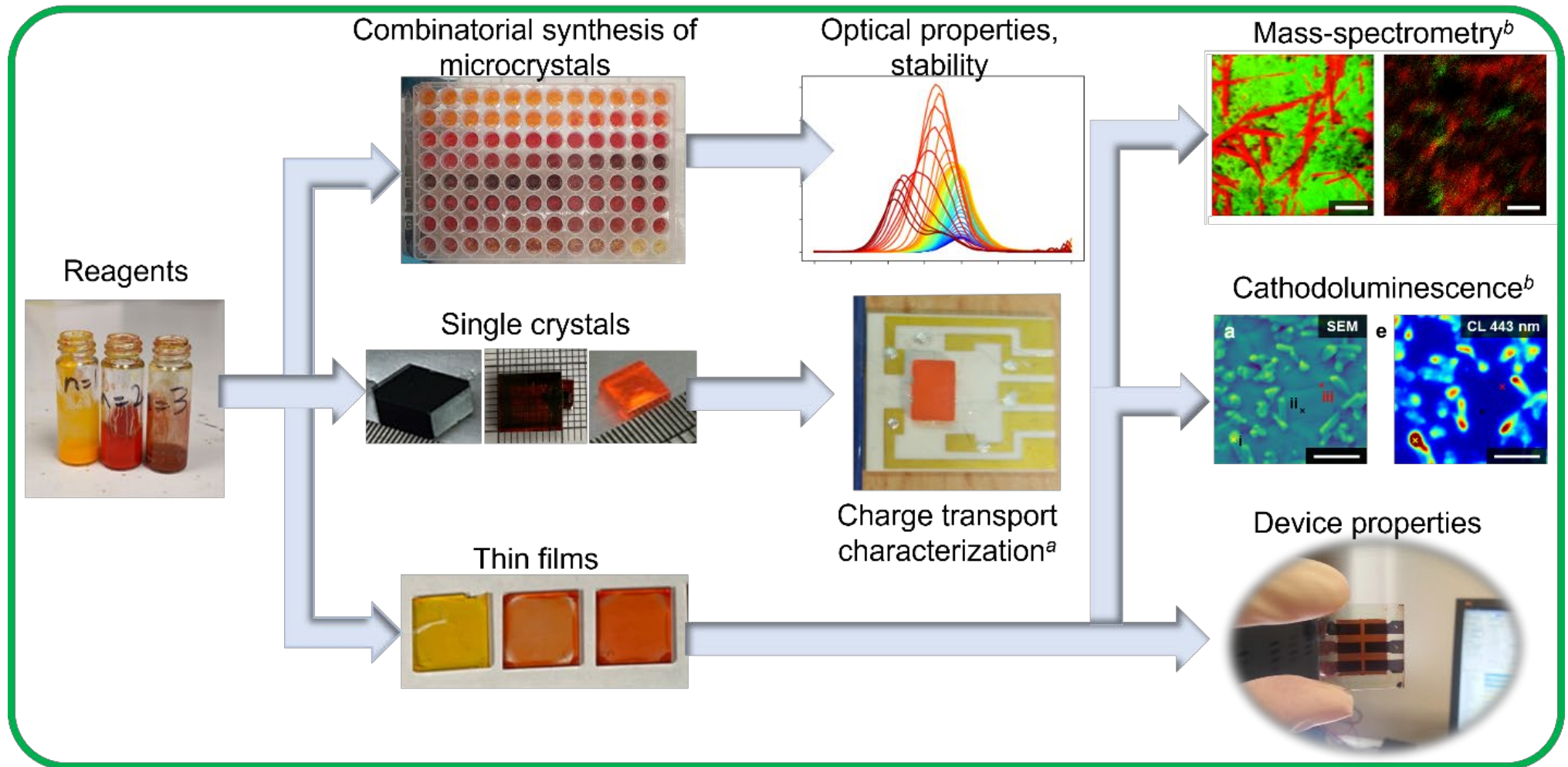
Lecture 02: History of ML and Scientific Data

Instructor: Sergei V. Kalinin

Why do we need machine learning?

To help us make decisions!

What is A Workflow?



- **Workflow:**
- Ideation, orchestration, implementation
- Domain specific language
- Dynamic planning: latencies and costs
- Reward and value functions

Designed in academia and adopted by industry

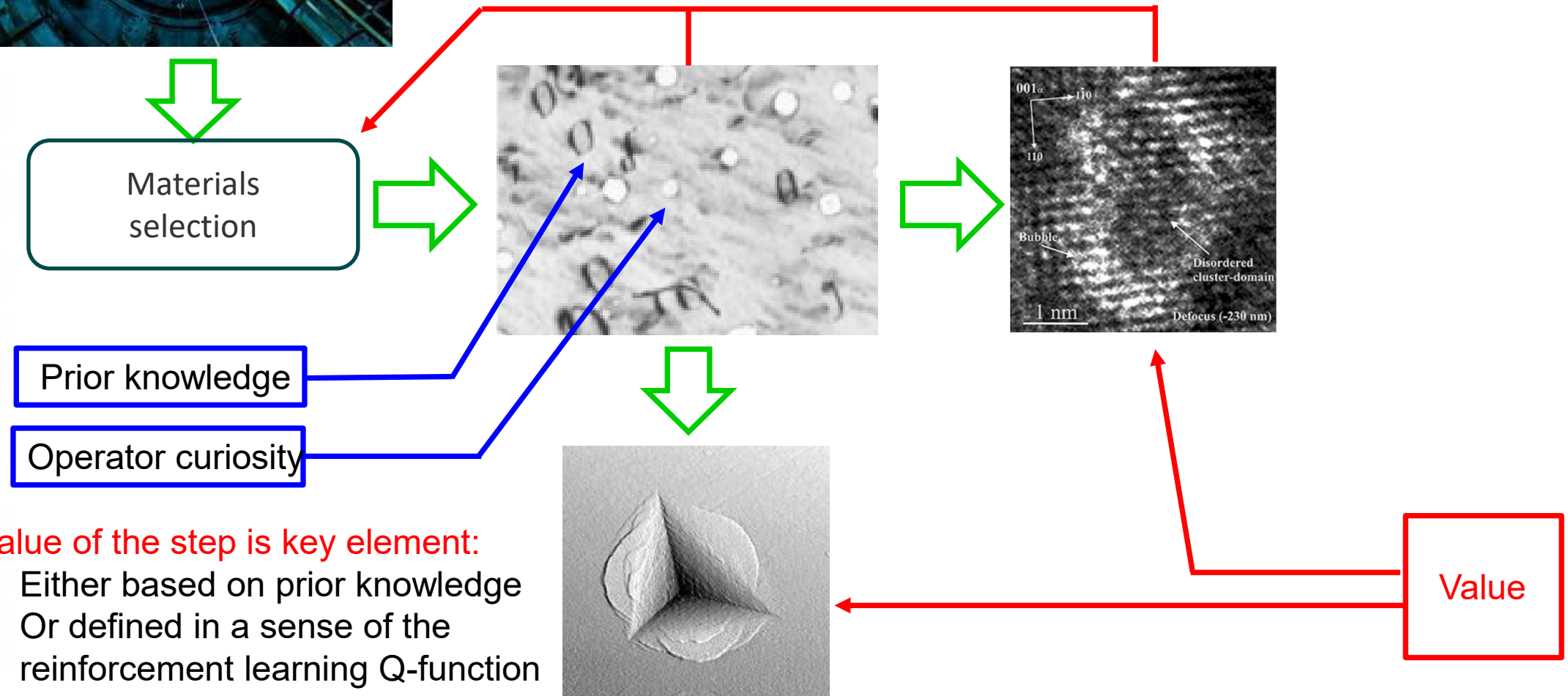
- Are they optimal?
- Can we design them better?
- Can they be changed dynamically?

Workflows for Nuclear Materials Design



Traditional experiment:

1. Always based on workflows
2. Ideated, orchestrated, and implemented by humans
3. The “gain of value” during the workflow implementation is uncertain

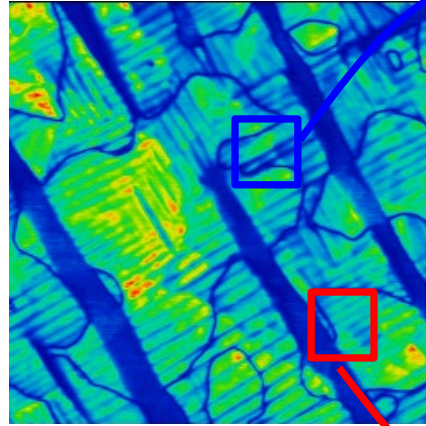


Workflows in Scanning Probe Microscopy

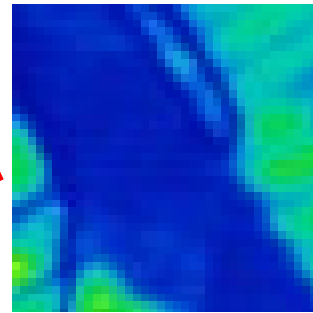
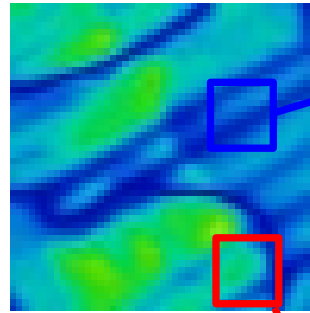
Workflow Plane

Workflow plane

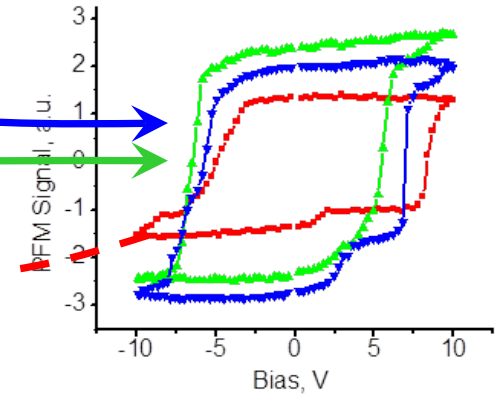
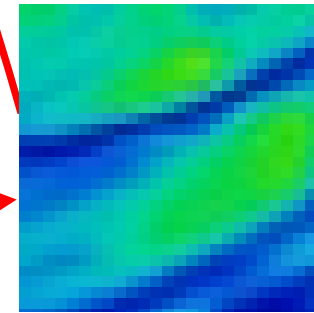
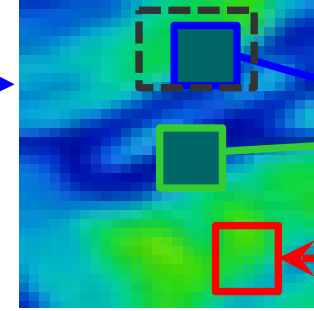
Overview scan



Zoom in



Zoom in

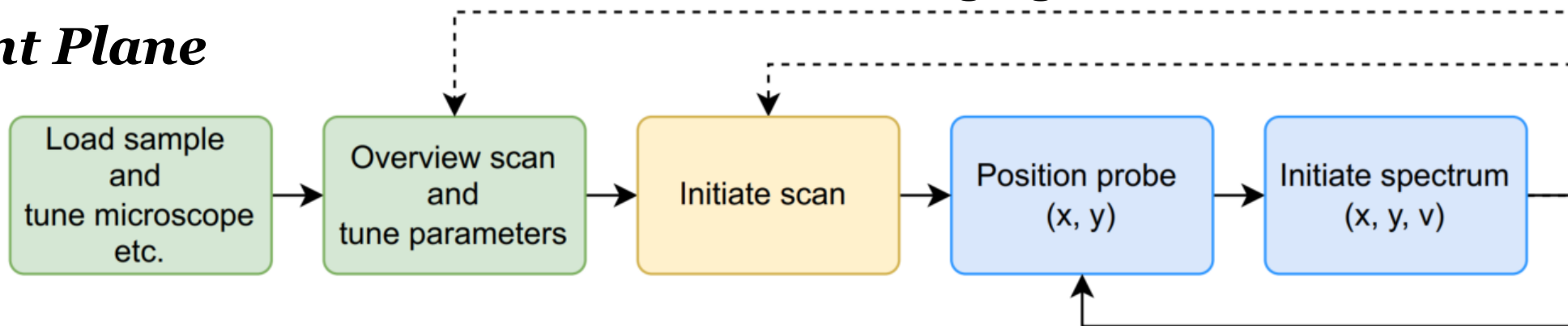


After acquisition analysis

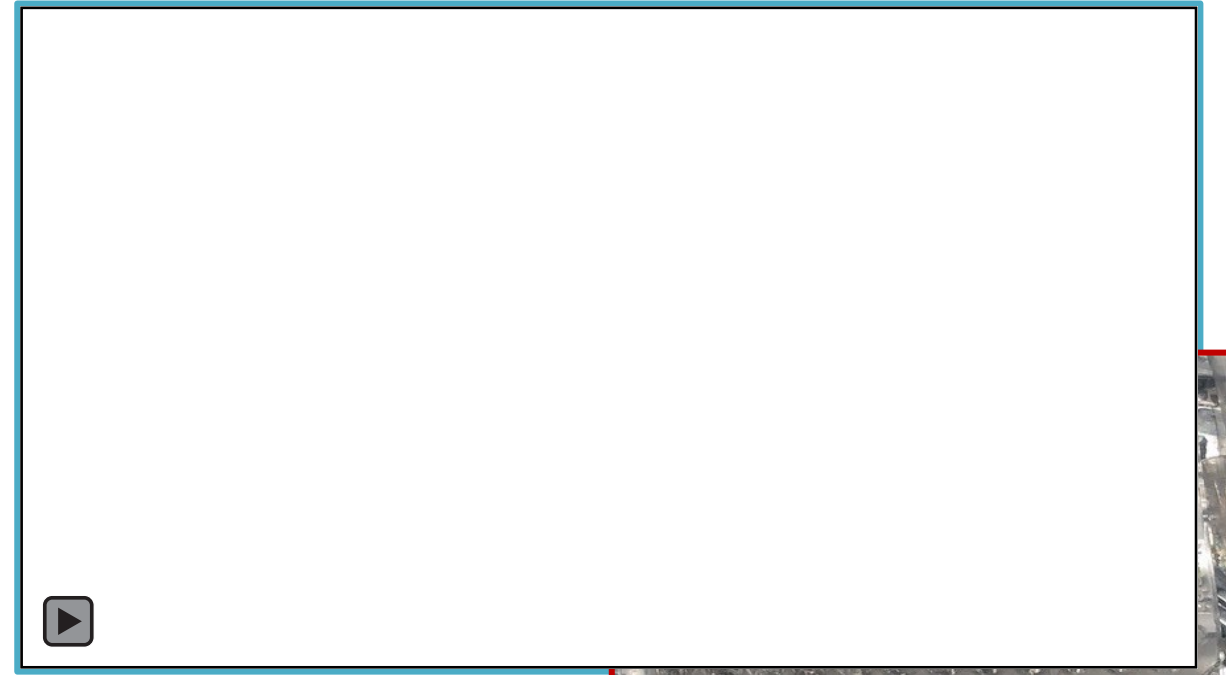
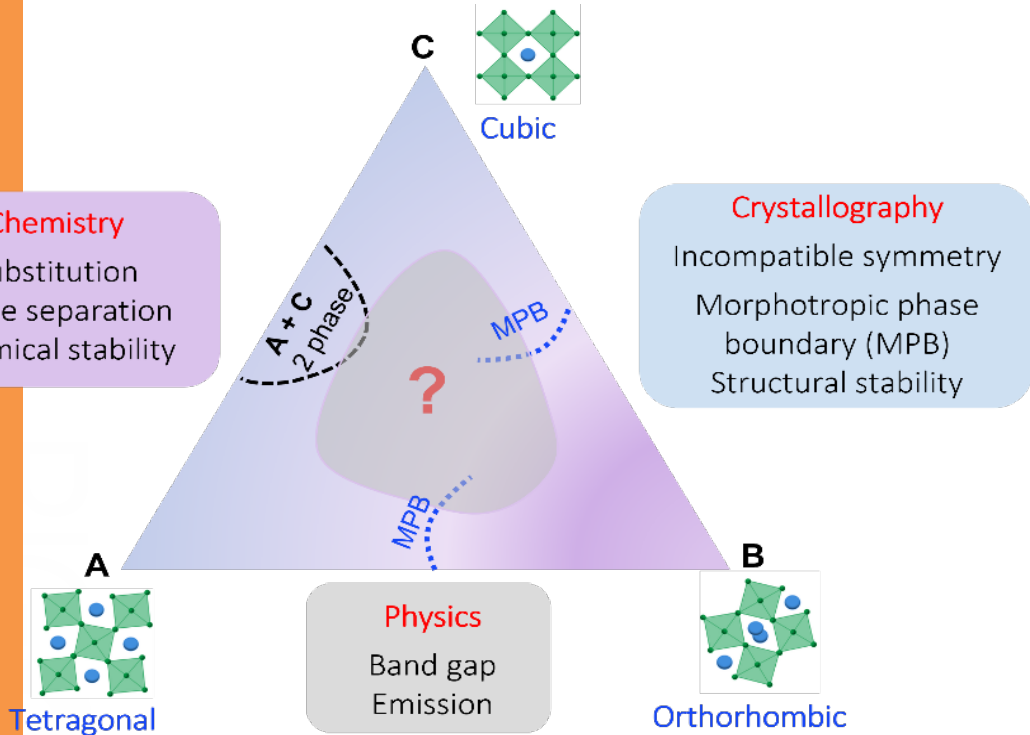
Instrument Plane

Instrument plane

Minimal instruction set control language

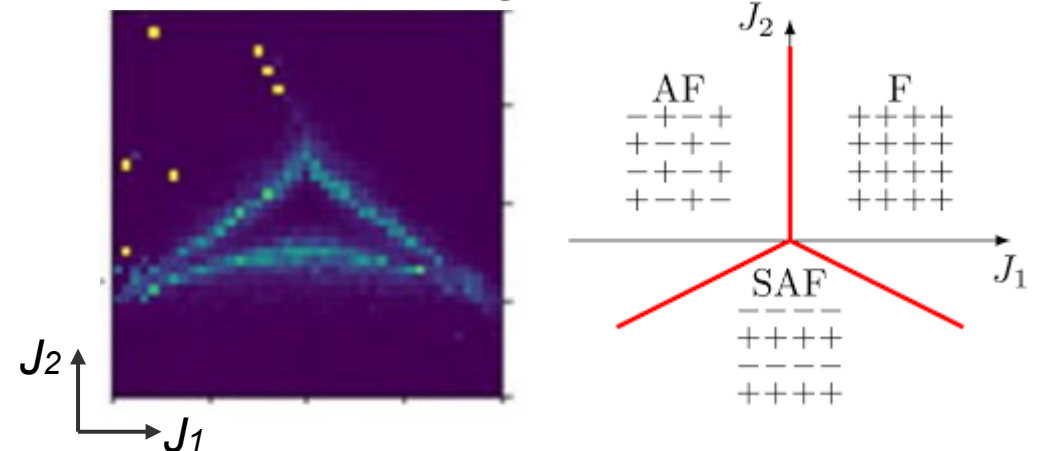


Why synthesis (or theory)?



- Automated synthesis in its simplest form requires some way to navigate phase diagrams
- In more complex form, processing space.
- Ideally, incorporate physical knowledge
- Similar problem - theory

Ising model



How we make decisions:

human + AI

How we execute decisions:

human + (automated) instrumentation

Making it real:

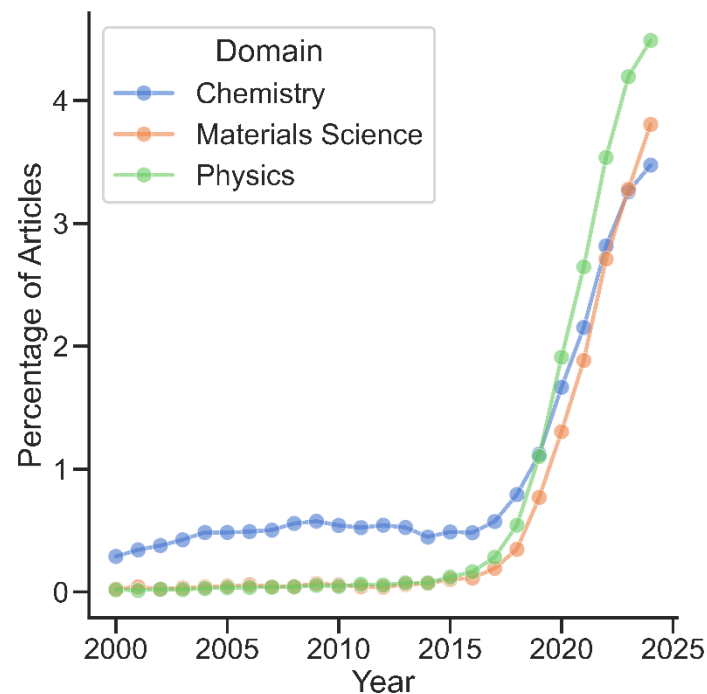
- What decisions can be executed?
- How fast can decisions be made?
- How fast can decisions be executed?
- How do we improve decisions?

Each hypothesis requires defining experimental action space

Curriculum and philosophy

ML Adoption in New Areas:

1. Solve extant problems that you can formulate (LLMs can help)
2. New ML opportunities: VAEs, game theory, decision theory, etc.
3. Building representations: there is no computation without representation
4. Building rewards, heuristics, and roll-outs: using ML for the real world problems



Analysis by B. Blaiszik, Argonne

Course Information

Faculty Contact Information:

Instructor: Prof. Sergei V. Kalinin,
Office: 314 IAMM
E-mail: sergei2@utk.edu
Teaching Assistant: Kamyar Barakati, kbarakat@vols.utk.edu

Instructor Availability:

Please don't hesitate to email me with updates, questions, or concerns. I will typically respond within 24 hours during the week and 48 hours on the weekend (unless I travel). I will notify you if I will be out of town and if connection issues may delay a response.

Meeting Time: 10:20 am - 11:10 am MWF, Ferris Hall 511

The lectures and materials will be posted on Canvas and at GitHub:
https://github.com/SergeiVKalinin/MSE_Fall_2026

Office Hours:

Friday 1:30 - 3:00 PM are open for 1:1 meetings to discuss any course related item. Please set up time via email.

Prerequisites

To be successful in this course you will need a general background in materials science. Python or similar programming experience, while not essential, will be extremely useful. Students without any prior programming experience should expect to spend extra time outside of class learning basic skills.

This and that

Learning Environment:

The class will be delivered as in-person lectures. The Jupyter notebooks, code libraries, and videos provided. Weekly programming exercises will be assigned via Google Colabs and those students wishing to interact with the instructor in person should attend office hours.

Use of ChatGPT:

Strongly encouraged both for programming and written assignments. However, the students have to be aware of the limitations of the generative models.

Grading & Policies:

- | | |
|--------------------------------|-----|
| • Homework assignments | 40% |
| • Mid-terms (2) | 30% |
| • Final Project & Presentation | 30% |

Reference Materials

I will provide copies of lecture notes, presentations, and Colabs on GitHub and Canvas. There is no specific textbook for the course and we will take material from a variety of sources including:

- Andrew Bird et al, *Python Workshop – Second Edition*,
<https://subscription.packtpub.com/book/programming/9781804610619/1>
- Sebastian Raschka, *Machine Learning with PyTorch and Scikit-Learn*,
<https://subscription.packtpub.com/book/data/9781801819312/1>
- Rowel Atienza, *Advanced Deep Learning with TensorFlow 2 and Keras - Second Edition*,
<https://www.packtpub.com/product/advanced-deep-learning-with-tensorflow-2-and-keras-second-edition/9781838821654>
- (Optional) Alaa Khamis, *Optimization Algorithms: AI Techniques for Design, Planning, and Control Problems*, <https://www.manning.com/books/optimization-algorithms>
- (Optional) Peter Norvig, *Artificial Intelligence: A Modern Approach*, Global Edition,
<https://www.amazon.com/Artificial-Intelligence-Modern-Approach-Global/dp/1292401133>

Homework 1:

- Create new Colab, <https://colab.google/>
- Explore NotebookLM, <https://notebooklm.google/>
- Chapter 1-4, Python Workshop.

Zooming out on history

K. Pearson, 1901

[559]

LIII. *On Lines and Planes of Closest Fit to Systems of Points in Space.* By KARL PEARSON, F.R.S., University College, London*.

(1) IN many physical, statistical, and biological investigations it is desirable to represent a system of points in plane, three, or higher dimensioned space by the "best-fitting" straight line or plane. Analytically this consists in taking

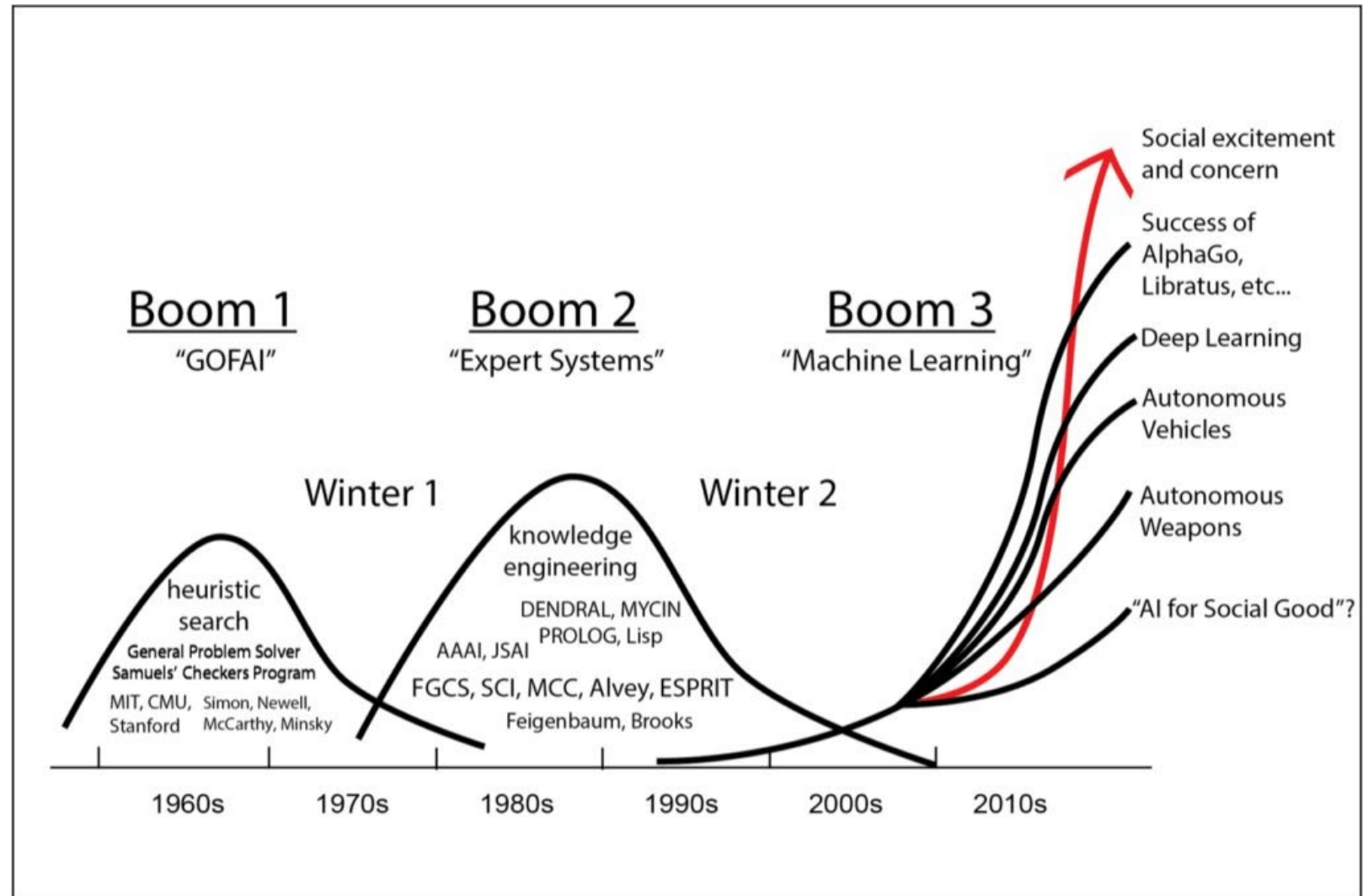
$$y = a_0 + a_1x, \text{ or } z = a_0 + a_1x + b_1y, \\ \text{or } z = a_0 + a_1x_1 + a_2x_2 + a_3x_3 + \dots + a_nx_n,$$

where $y, z, x_1, x_2, \dots, x_n$ are variables, and determining the "best" values for the constants $a_0, a_1, b_1, a_0, a_1, a_2, a_3, \dots, a_n$ in relation to the observed corresponding values of the variables. In nearly all the cases dealt with in the text-books of least squares, the variables on the right of our equations are treated as the independent, those on the left as the dependent variables. The result of this treatment is that we get one straight line or plane if we treat some one variable as independent, and a quite different one if we treat another variable as the independent variable. There is no paradox about this; it is, in fact, an easily understood and most important feature of the theory of a system of correlated variables. The most probable value of y for a given value of x , say, is not given by the same relation as the most probable value of x for a given value of y . Or, to take a concrete example, the most probable stature of a man with a given length of leg l being s , the most probable length of leg for a man of stature s will not be l . The "best-fitting" lines and planes for the cases of z up to n variables for a correlated system are given in my memoir on regression†. They depend upon a determination of the means, standard-deviations, and correlation-coefficients of the system. In such cases the values of the independent variables are supposed to be accurately known, and the probable value of the dependent variable is ascertained.

(2) In many cases of physics and biology, however, the "independent" variable is subject to just as much deviation or error as the "dependent" variable. We do not, for example, know x accurately and then proceed to find y , but both x and y are found by experiment or observation. We observe x and y and seek for a unique functional relation between them. Men of given stature may have a variety

* Communicated by the Author.

† Phil. Trans. vol. clxxxvii. A, pp. 301 *et seq.*



<https://jaylatta.net/history-of-ai-from-winter-to-winter/>

Visions of the Future



AI Apocalypse: 80% of Projects Crash and Burn, Billions Wasted says RAND Report

📅 August 19, 2024 👤 Vernon Keenan 📁 Industry Analysis 💬 0 Comments

A

new [RAND Corporation](#) report reveals the sobering reality behind artificial intelligence (AI) projects: despite the hype, most of them fail. The study, based on interviews with 65 experienced data scientists and engineers, exposes the root causes of these failures and offers a roadmap for success.

Upcoming Events

AUG 8:00 am - 9:00 am
22 Transform Your Salesforce
DevOps Tooling and
Practice with AI-Driven
OpsBridge Frameworks

[View Calendar](#)

[GET FREE EMAIL UPDATES](#)

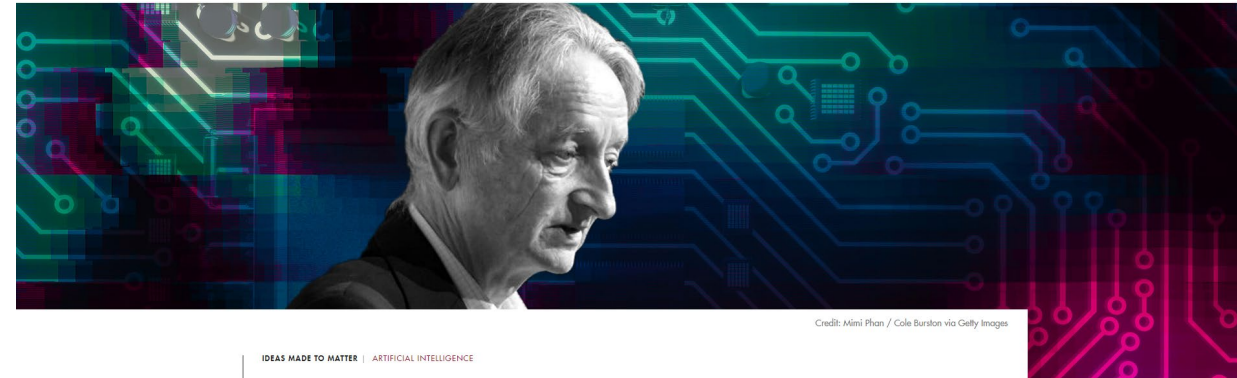
Taking the Human Out of the Loop: A Review of Bayesian Optimization

Citation

Shahriari, Bobak, Kevin Swersky, Ziyu Wang, Ryan P. Adams, and Nando de Freitas. 2016. "Taking the Human Out of the Loop: A Review of Bayesian Optimization." *Proc. IEEE* 104 (1) (January): 148–175. doi:10.1109/jproc.2015.2494218.

Published Version

doi:10.1109/JPROC.2015.2494218



Credit: Hini Phan / Cole Burston via Getty Images

IDEAS MADE TO MATTER | ARTIFICIAL INTELLIGENCE

Why neural net pioneer Geoffrey Hinton is sounding the alarm on AI

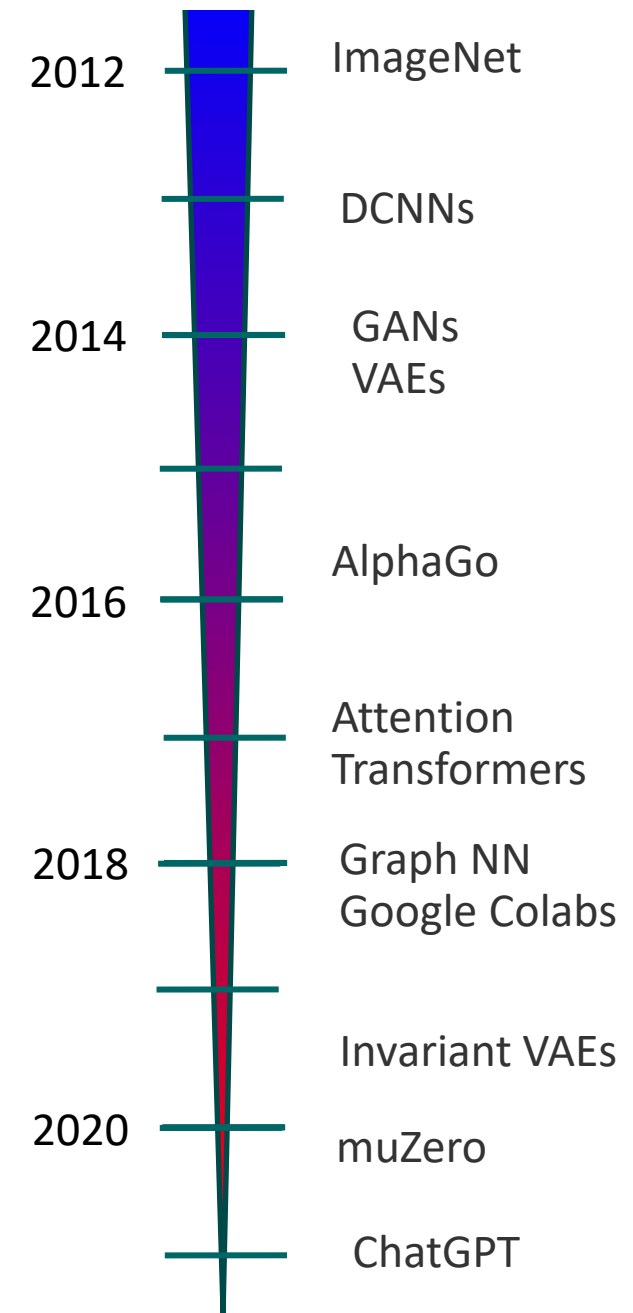
ML of the last decade

- Last decade has experienced an explosive growth of machine learning and artificial intelligence applications
- These developments have spanned areas from computer vision to medicine to autonomous systems and games
- However, the progress and impact as applied to experimental physical sciences has been minimal....

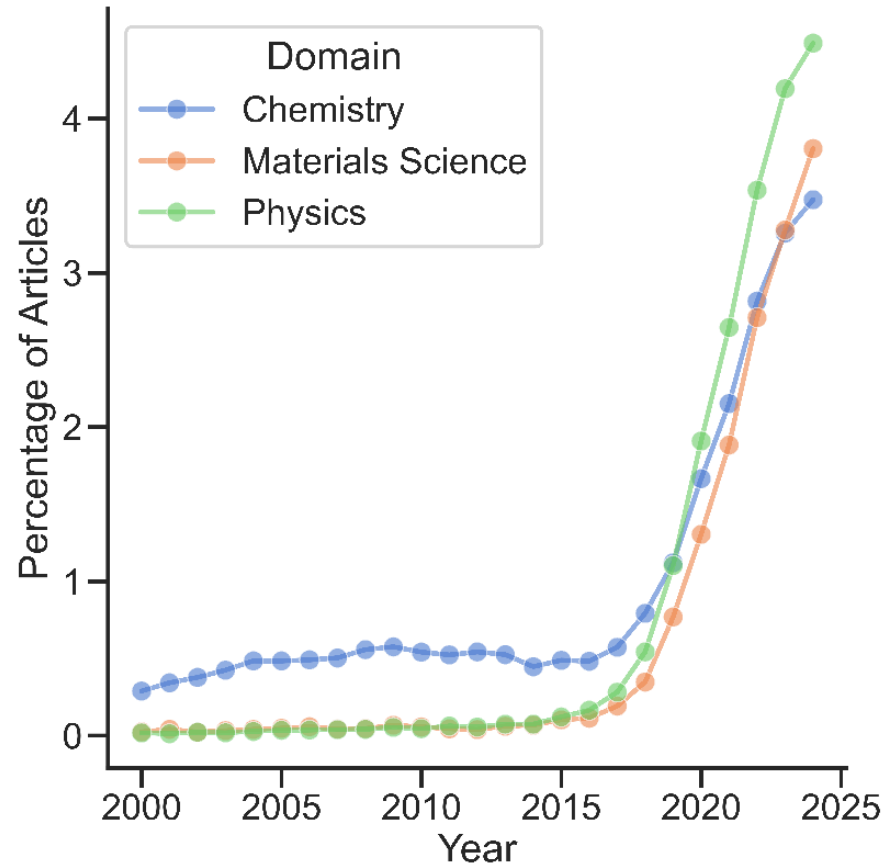
Why is it difficult?

- Requires domain expertise and domain-specific goals
- Deeply causal and hypothesis drive nature of domain sciences
- No single answer: culture, not a method
- Infrastructure, open code, open data
- **Most important:** active nature of scientific process

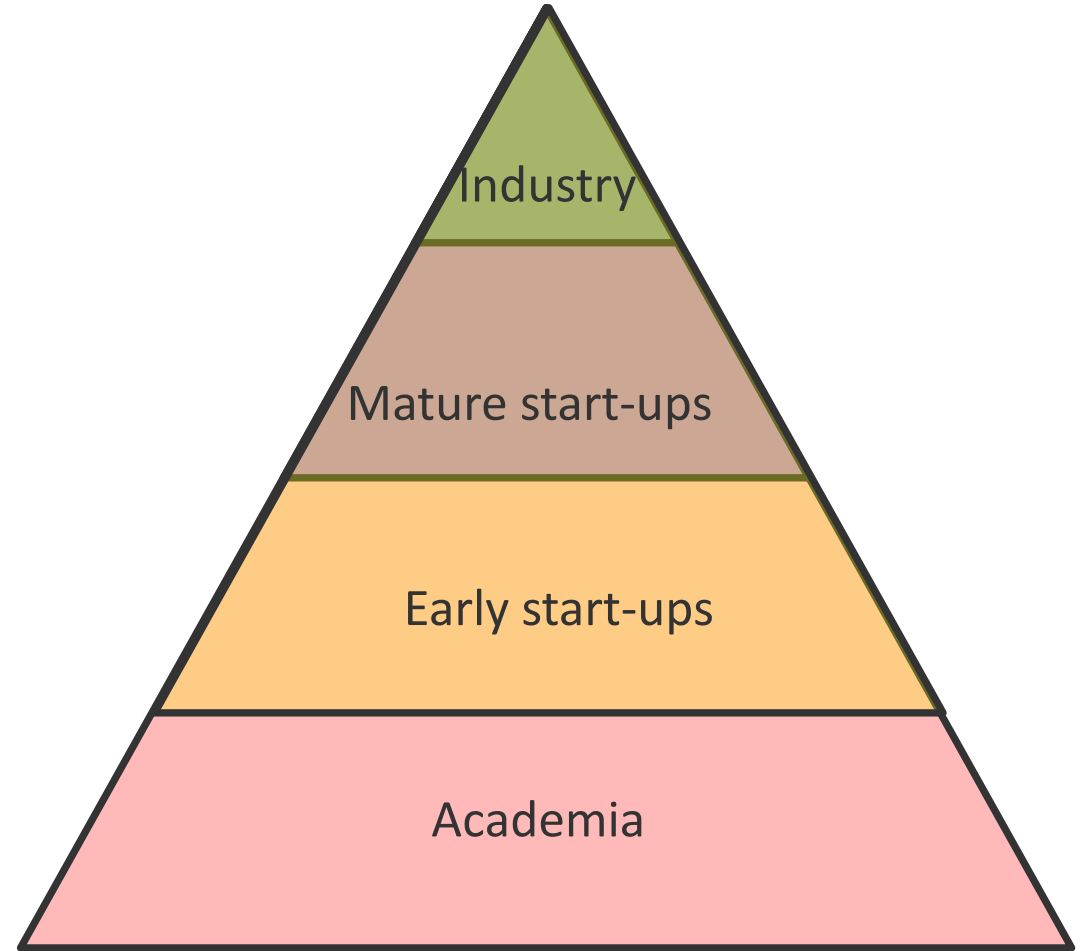
Microsoft: GitHub
Meta: Open Catalyst,
Meta: Papers with Code
Toyota: TRI
Google: AlphaFold
NVIDIA: protein folding



ML in Domain Sciences



Analysis by B. Blaiszik, Argonne



- The rapid adoption of ML in domain sciences and industrial R&D is a very recent trend
- Technologies and workforce emerge from academia into industry
- We can estimate potential growth rates comparing to cloud computing 15 - 20 years ago

“Eras” of ML in Industry

- **Before 2000:** It's all about IT (dotcoms, Amazon, etc)
 - **2000 - 2010:** It's all about collecting and searching data (Facebook, Google, Uber)
 - **2010 – 2020:** What do we learn from data (correlative era)
 - **2020 – now:** Physics is the new data
-
- Classical machine learning is underpinned by the existence of the **large static data sets** – from MNIST to emerging medical, bio, faces, etc.
 - Real world problems are associated with the large distribution shifts, often small data sets, and presence of uncontrollable exogenous factors
 - Also, real world problems are often **active learning**: we interrogate the data generation process and provide feedback, not deal with static data sets
 - However, we often have extensive **prior knowledge** of past data, **physical laws** generalizing them, **human heuristics**, and strong set of inferential biases

ML for real-world applications is different!



More than 1.5 \$B VC
investment over the
last year!

Prometheus: \$6B

- Lila Scientific
- Periodic Labs
- Radical AI
- Phase Tree
- Aionics
- Eionics
- Atinary
- Google
- Amazon
- ...

The “Multiple-Sample Concept” in Materials Research: Synthesis, Compositional Analysis and Testing of Entire Multicomponent Systems

J. J. HANAK

RCA Laboratories, Princeton, New Jersey, USA

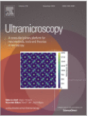
**Combinatorial Synthesis and Reactivity Screening of
Electro-Oxidation Catalyst Gradients 171**
Shrisudersan Jayaraman and Andrew C. Hillier

**Synthesis and Screening of Materials Libraries of Buried
Compound Semiconductors by Ion Beam Implantation**
I. Großhans, H. Karl, and B. Stritzker

**Controlled Deposition of Wedge-Shaped Profiles for
Thin (~10Å) Layers by Pulsed Laser Deposition.....**
H.M. Christen and I. Ohkubo



Ultramicroscopy
Volume 59, Issues 1-4, July 1995, Pages 229-239



Quantification

Analyzing line scan EELS data with neural pattern recognition

C. Gatts¹, G. Duscher , H. Mülleijans, M. Rühle

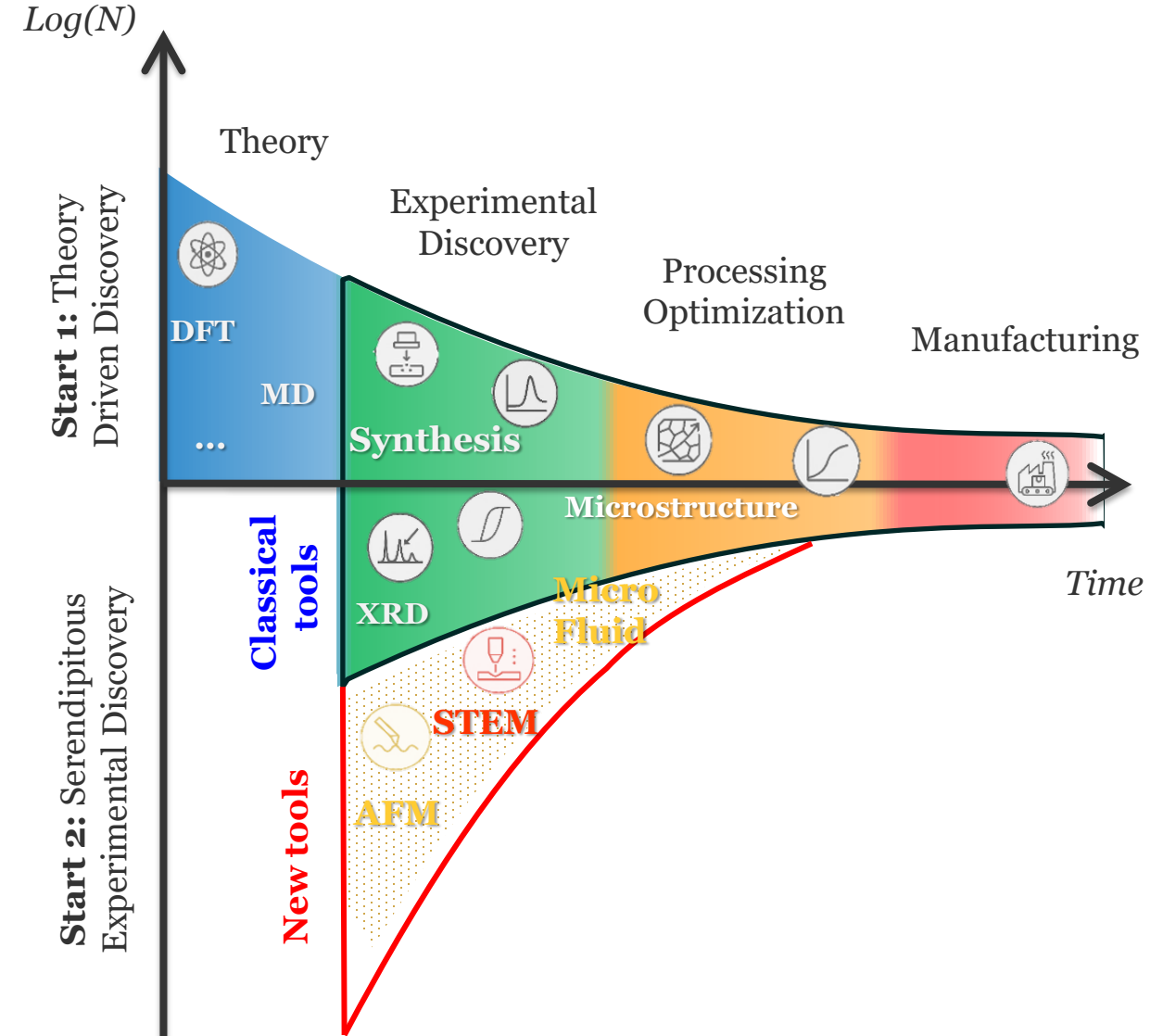
SYMPOSIUM S

Combinatorial and Artificial Intelligence Methods in Materials Science

November 26 – 29, 2001

“Those who cannot remember the past are condemned to repeat it.”
— George Santayana, *The Life of Reason: Reason in Common Sense* (1905)

1. **Close the loop:** To accelerate materials discovery, we have to close the loop from synthesis to characterization and model update
2. **Explore smarter:** ML/AI methods along with the predictive theory allow better initial choices; however, most materials are discovered serendipitously even now
3. **New tool paradigms:** Characterization tools are often limited by fundamental physics: we need different tool paradigms!



Types of Machine Learning

Supervised (inductive) learning

- Given: training data + desired outputs (labels)

• **Unsupervised learning**

- Given: training data (without desired outputs)

• **Semi-supervised learning**

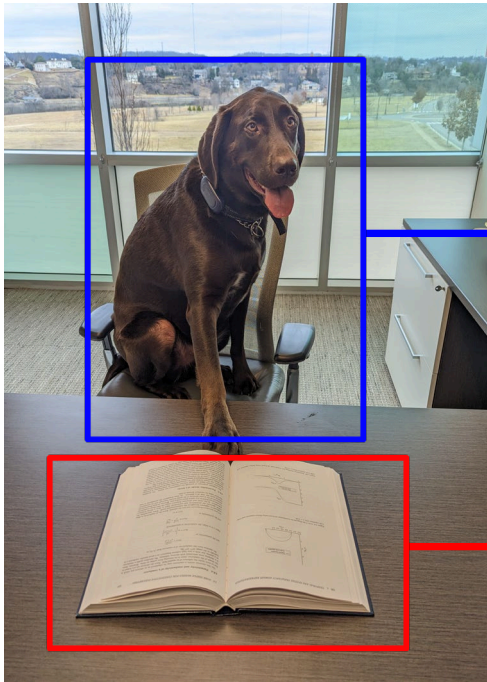
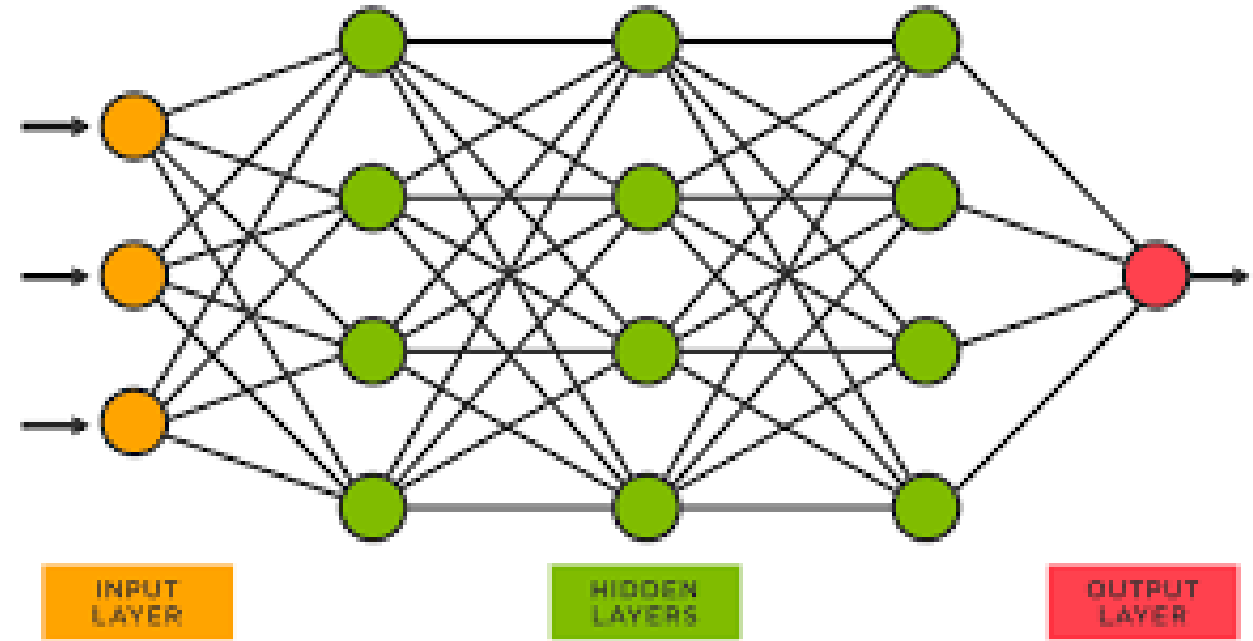
- Given: training data + a few desired outputs

• **Reinforcement learning**

- Rewards from sequence of actions

Supervised Machine Learning

- Regression
- Classification
- Semantic segmentation
- Instance segmentation
- ...



Dog

Book

Classification

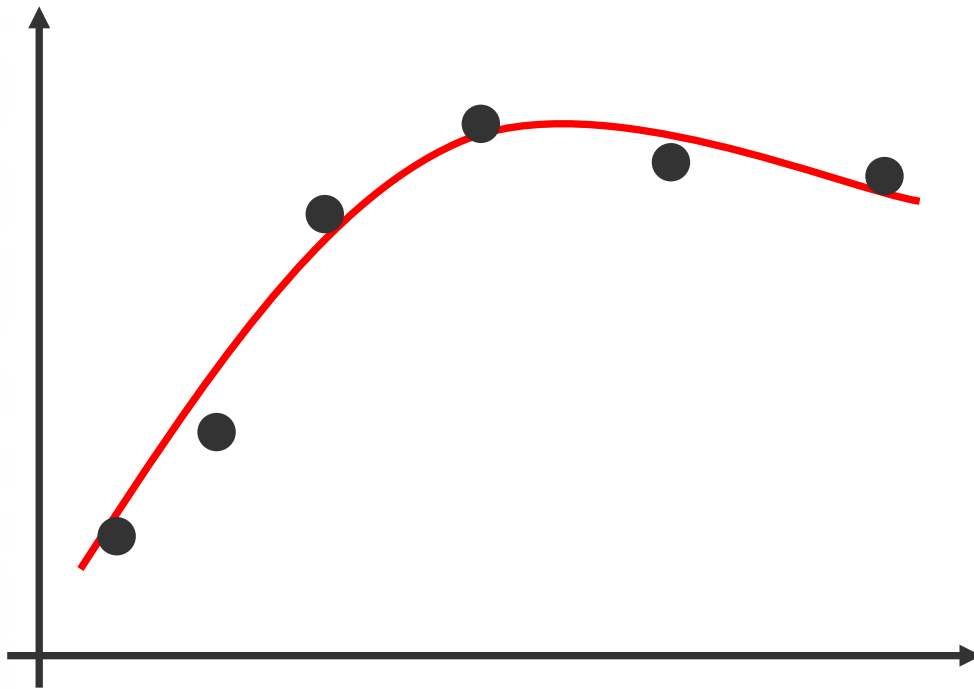
- Given $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$
- Learn a function $f(x)$ to predict y given x
- If y is categorical == classification

Application	Input Data	Classification
Medical Diagnosis	Noninvasive tests	Results from invasive measurements
Optical Character Recognition	Scanned bitmaps	Letter A-Z and digits 0-9
Protein Folding	Amino acid sequence	Protein shape (helices, loops, sheets)
Materials Discovery	Composition	Metal/Semiconducotr
Research Paper Acceptance	Words in paper title	Paper accepted or rejected

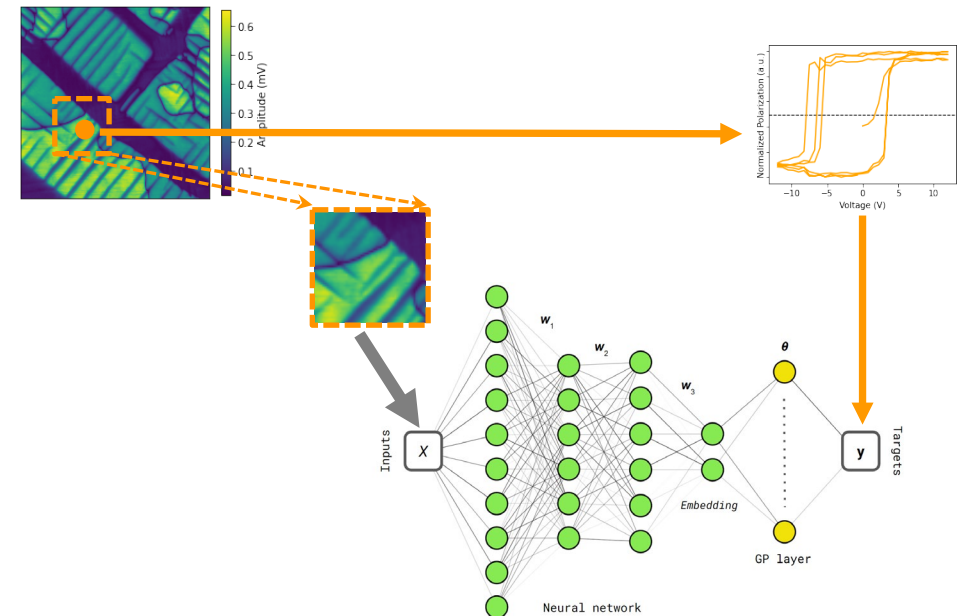
Regression

- Given $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$
- Learn a function $f(x)$ to predict y given x
- y is real-valued == regression

Simple regression: $\mathbb{R}^1 \rightarrow \mathbb{R}^1$

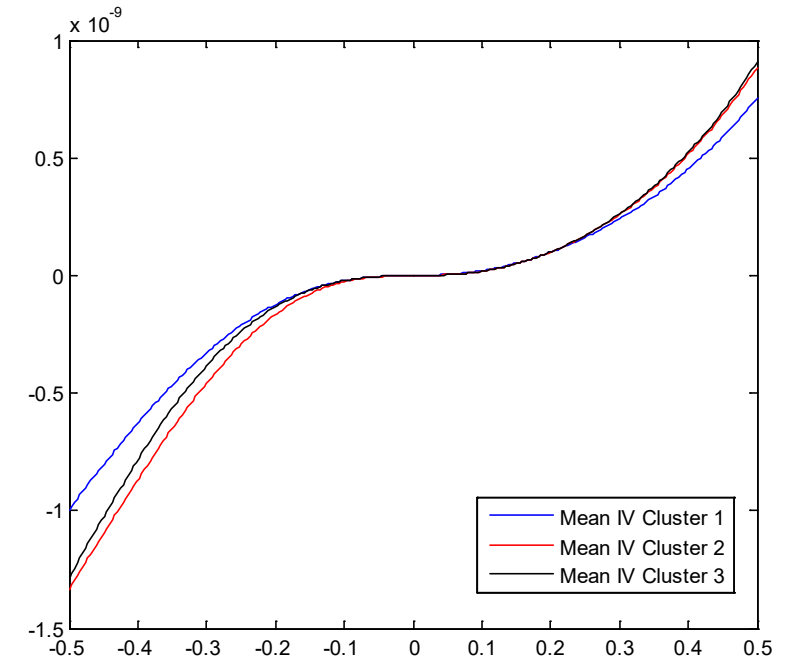
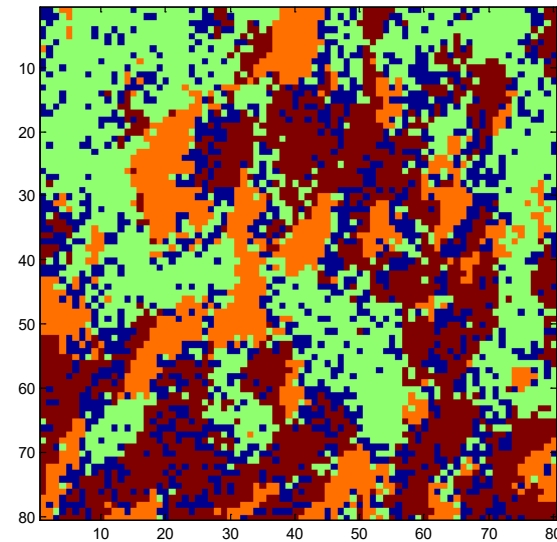
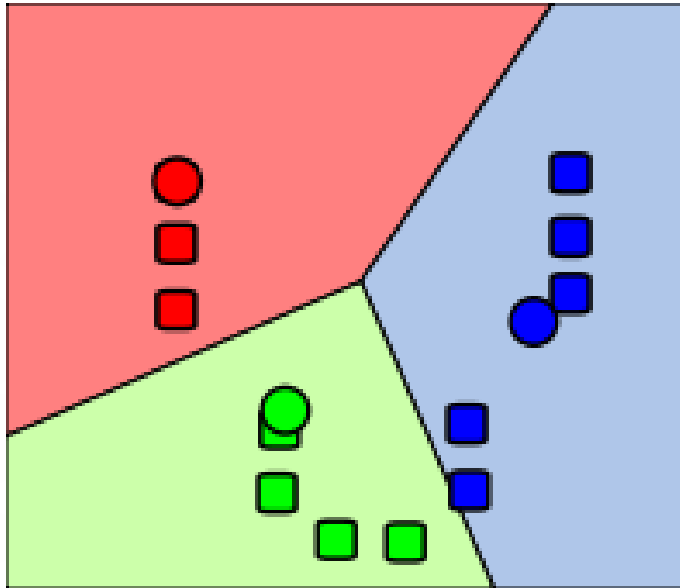


Not so simple regression



Unsupervised Learning

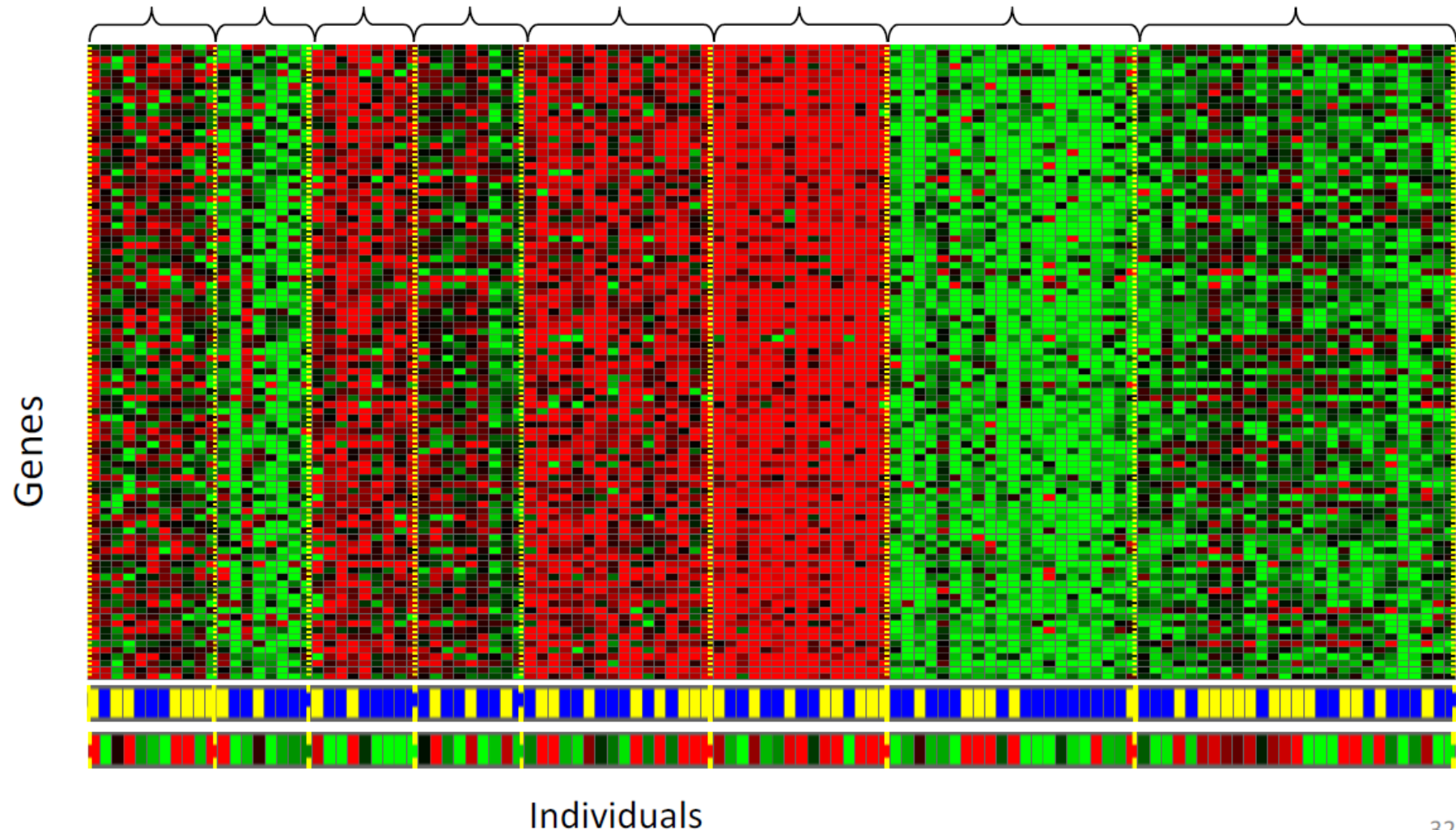
- Given $x_1, x_2, \dots, x_n$ (without labels)
- Output hidden structure behind the x 's
- Example: clustering



M. ZIATDINOV, A. MAKSOV, L. LI, A. SEFAT, P. MAKSYMОВYCH, and S.V. KALININ, *Deep data mining in a real space: Separation of intertwined electronic responses in a lightly-doped BaFe₂As₂*, Nanotechnology **27**, 475706 (2016).

Unsupervised Learning

Genomics application: group individuals by genetic similarity

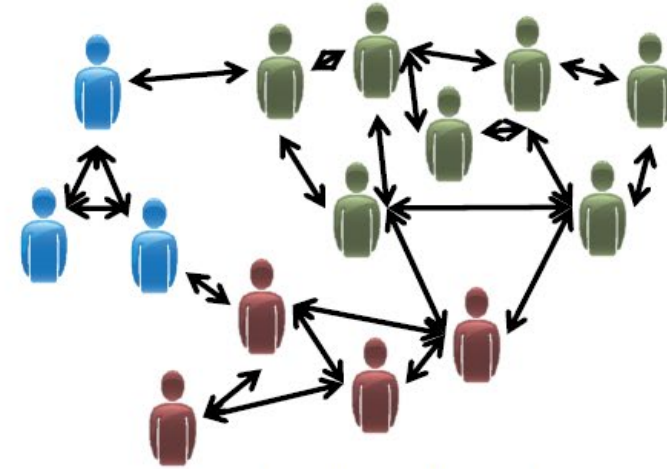


[Source: Daphne Koller]

Unsupervised Learning



Organize computing clusters



Social network analysis



Market segmentation

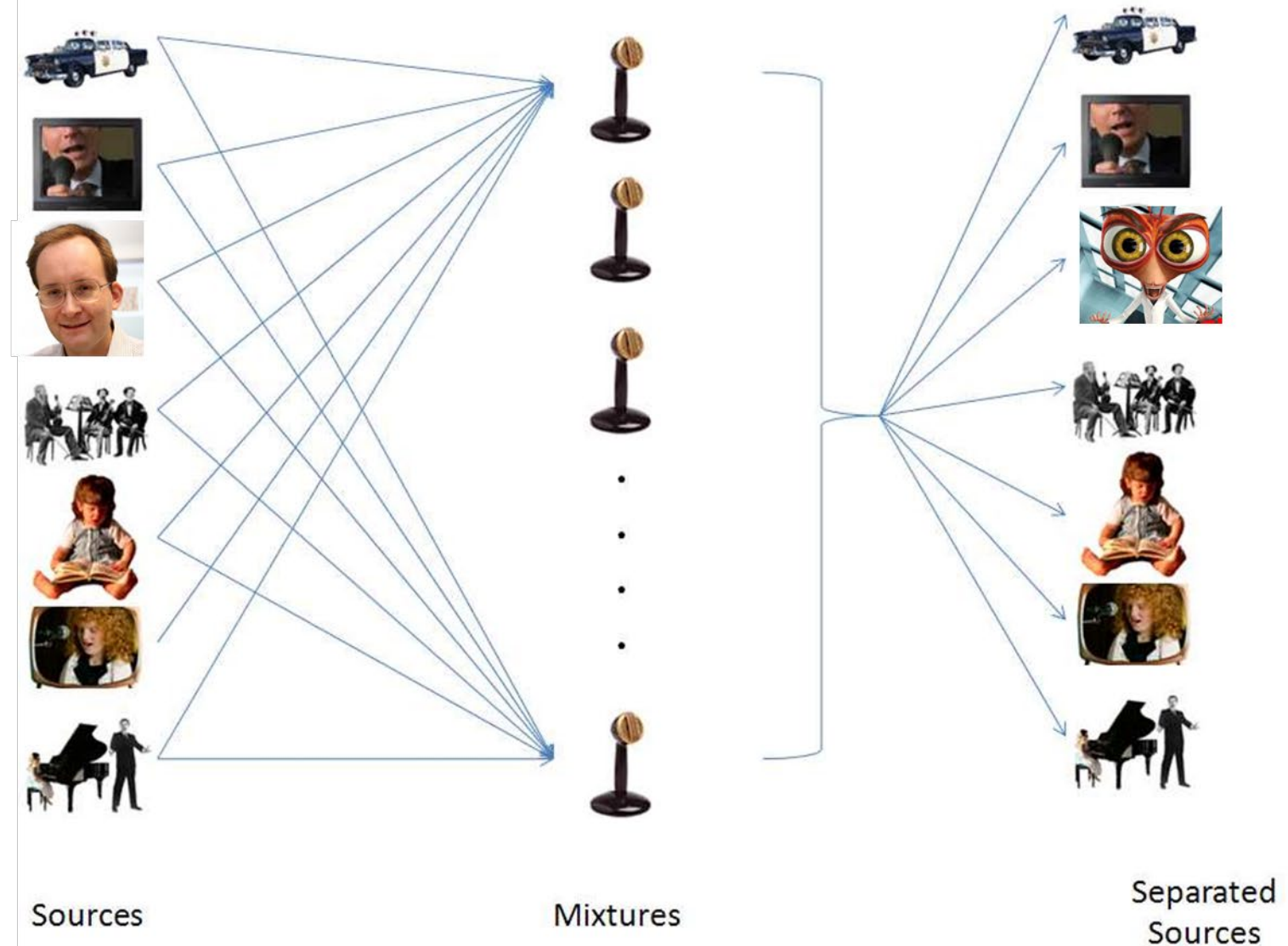
Slide credit: Andrew Ng



Astronomical data analysis

Unsupervised Learning

Number of signals are being produced simultaneously; with the objective of separating and following each source separately

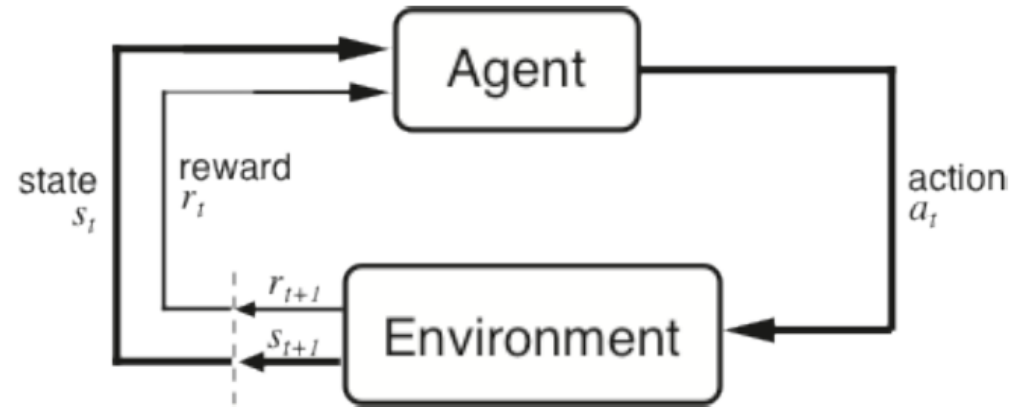


Reinforcement Learning

Given a sequence of **states** and **actions** with (delayed) **rewards**, output a policy, i.e. a mapping from states to actions that tells you what to do in a given state

- Examples:
 - Credit assignment problem
 - Game playing
 - Robot in a maze
 - Balance a pole on your hand

RL: Agent and Environment



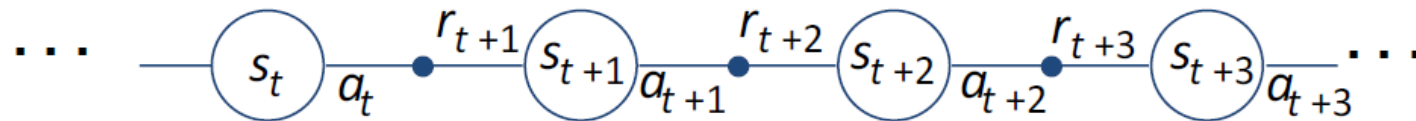
Agent and environment interact at discrete time steps : $t = 0, 1, 2, K$

Agent observes state at step t : $s_t \in S$

produces action at step t : $a_t \in A(s_t)$

gets resulting reward : $r_{t+1} \in \mathfrak{R}$

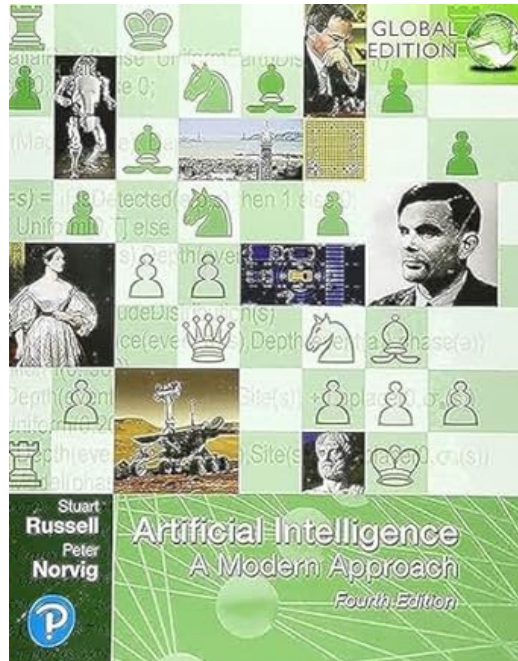
and resulting next state : s_{t+1}



Reinforcement Learning in Action



<https://www.youtube.com/watch?v=GtYIVxv0py8>



Somewhat remarkably, almost all AI research until very recently has assumed that the performance measure can be exactly and correctly specified in the form of utility or reward function

Reinforcement Learning Applications

Chemical Synthesis and Drug Discovery



M. Ahmadi lab
UTK MSE



Cloud Laboratories



Emerald Cloud Lab,
SF and CMU

